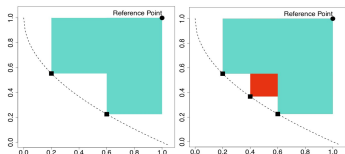
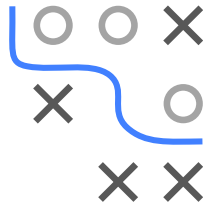


Optimization in Machine Learning

Multi-criteria Optimization

Bayesian Optimization



Learning goals

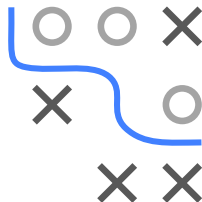
- Single-criteria recap
- Multi-criteria introduction
- Scalarization vs. HV-based methods

RECAP: BAYESIAN OPTIMIZATION

Advantages of BO

- Sample efficient
- Can handle noise
- Native incorporation of priors
- Does not require gradients
- Theoretical guarantees

We will now extend BO to multiple cost functions.



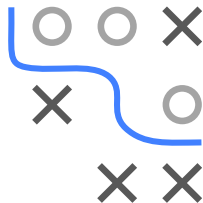
RECAP: BAYESIAN OPTIMIZATION

Algorithm Bayesian optimization loop

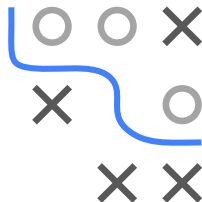
Require: Search space $\mathcal{S}$, cost function f , acquisition function u , predictive model $\hat{f}$, maximal number of function evaluations T

Ensure: Best configuration $\hat{\mathbf{x}}$ (based on $\mathcal{D}$ or $\hat{f}$)

- 1: Initialize data $\mathcal{D}^{(0)}$ with initial observations
 - 2: **for** $t \leftarrow 1$ to T **do**
 - 3: Fit predictive model $\hat{f}^{(t)}$ on $\mathcal{D}^{(t-1)}$
 - 4: Select next query point:
 - 5: $\mathbf{x}^{(t)} \in \arg \max_{\mathbf{x} \in \mathcal{S}} u(\mathbf{x}; \mathcal{D}^{(t-1)}, \hat{f}^{(t)})$
 - 6: Query $f(\mathbf{x}^{(t)})$
 - 7: Update data:
 - 8: $\mathcal{D}^{(t)} \leftarrow \mathcal{D}^{(t-1)} \cup \{\langle \mathbf{x}^{(t)}, f(\mathbf{x}^{(t)}) \rangle\}$
 - 9: **end for**
-



MULTI-CRITERIA BAYESIAN OPTIMIZATION



Goal: Extend Bayesian optimization to multiple cost functions

$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x}) \quad \Leftrightarrow \quad \min_{\mathbf{x} \in \mathcal{S}} (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x})).$$

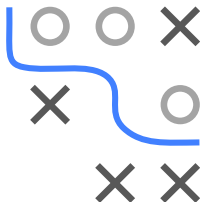
There are two basic approaches:

- 1 Simplify the problem by scalarizing the cost functions, or
- 2 define acquisition functions for multiple cost functions.

SCALARIZATION

Idea: Aggregate all cost functions

$$\min_{\mathbf{x} \in \mathcal{S}} \sum_{i=1}^m w_i f_i(\mathbf{x}) \quad \text{with} \quad w_i \geq 0$$

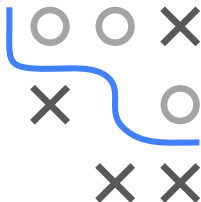


- **Obvious problem:** How to choose $w_1, \dots, w_m$?
 - Expert knowledge?
 - Systematic variation?
 - Random variation?
- If expert knowledge is not available a priori, we need to ensure that different trade-offs between cost functions are explored.
- Simplifies multi-criteria optimization problem to single-objective
→ standard Bayesian optimization can be used without adapting the general algorithm.

SCALARIZATION: PAREGO Knowles 2006

Scalarize the cost functions using the augmented Chebyshev norm / achievement function

$$f = \max_{i=1,\dots,m} (w_i f_i(\mathbf{x})) + \rho \sum_{i=1}^m w_i f_i(\mathbf{x}),$$



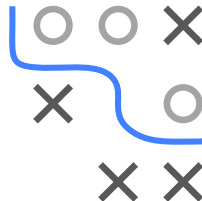
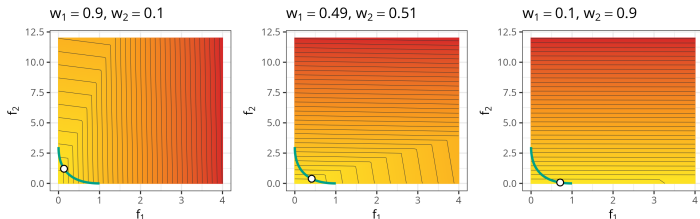
- The weights $w \in W$ are drawn from

$$W = \left\{ w = (w_1, \dots, w_m) \mid \sum_{i=1}^m w_i = 1, w_i = \frac{l}{s}, l \in \{0, \dots, s\} \right\},$$

with $|W| = \binom{s+m-1}{m-1}$.

- New weights are drawn in every BO iteration.
- ρ is a small parameter (e.g. 0.05).
- s controls how many distinct weight vectors exist.

WHY THE CHEBYSHEV NORM?



$$f = \max_{i=1,\dots,m} (w_i f_i(\mathbf{x})) + \rho \sum_{i=1}^m w_i f_i(\mathbf{x}),$$

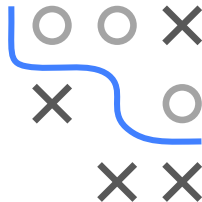
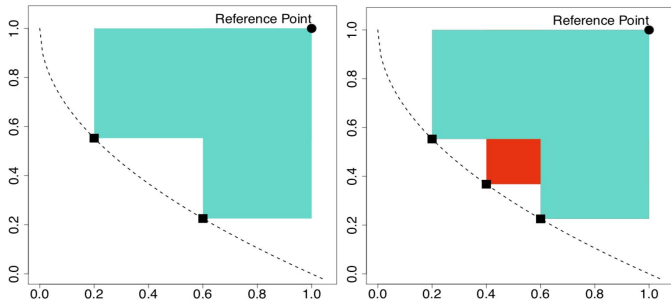
- The norm consists of two components:
 - $\max_i (w_i f_i(\mathbf{x}))$ considers only the maximum weighted cost.
 - $\sum_i w_i f_i(\mathbf{x})$ is the weighted sum of all costs.
- ρ controls the trade-off between these parts.
- By randomizing the weights w in each iteration (and keeping ρ small), extreme points in single cost functions can be explored.
- One can prove that every solution of this scalarized problem is Pareto-optimal.

PAREGO ALGORITHM

Algorithm ParEGO loop

HV-BASED ACQUISITION FUNCTIONS

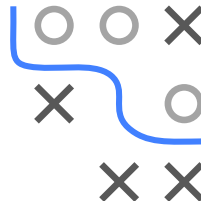
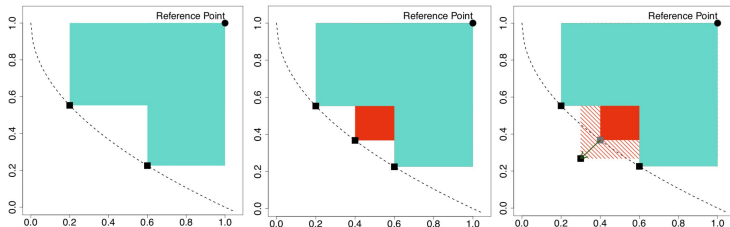
Idea: Define acquisition function that directly models the contribution to the dominated hypervolume (HV): $\max(0, S(\mathcal{P} \cup \mathbf{x}, R) - S(\mathcal{P}, R))$



- Fit m single-objective surrogate models $\hat{f}_1, \dots, \hat{f}_m$.
- Acquisition function uses all surrogate models simultaneously.
- Still a single-objective optimization problem (of acquisition itself).

S-METRIC SELECTION-BASED EGO

Using the Lower Confidence Bound $u_{\text{LCB},1}(\mathbf{x}), \dots, u_{\text{LCB},m}(\mathbf{x})$, one can form an *optimistic* estimate of the hypervolume contribution:



Problem: HV-contribution can be zero across large regions where the lower confidence bound is already dominated.

- Large zero-plateaus make optimization of the acquisition harder.
- SMS-EGO adds an adaptive penalty in such dominated regions.

This method is referred to as [Ponweiser et al. 2008](#).

FURTHER HV ACQUISITION FUNCTIONS

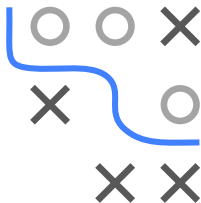
Expected Hypervolume Improvement (EHI) ► Yang et al. 2019:

$$u_{EI, \mathcal{H}}(\mathbf{x}) = \int_{-\infty}^{\infty} p(f \mid \mathbf{x}) \times [S(\mathcal{P} \cup \mathbf{x}, R) - S(\mathcal{P}, R)] df$$

- Direct extension of u_{EI} to the hypervolume criterion.
- $p(f \mid \mathbf{x})$ is the joint predictive density of all surrogate models at $\mathbf{x}$.
- With independent GPs per objective, this factorizes over m univariate normals.
- Exact integral feasible for $m \leq 3$, else simulation-based.

Further HV-based acquisitions:

- **Stepwise Uncertainty Reduction (SUR)** w.r.t. the probability of improvement.
- **Expected Maximin Improvement (EMI)** w.r.t. the ϵ -indicator.



HYPERVOLUME-BASED BO ALGORITHM

Algorithm Hypervolume-based Bayesian optimization loop

Require: Search space $\mathcal{S}$, cost function f , acquisition function u , predictive model(s) $\hat{f}_1, \dots, \hat{f}_m$, maximal number of function evaluations T

Ensure: Best configuration $\hat{\mathbf{x}}$

- 1: Initialize data $\mathcal{D}^{(0)}$ with initial observations
 - 2: **for** $t = 1$ to T **do**
 - 3: Fit predictive models $\hat{f}_1^{(t)}, \dots, \hat{f}_m^{(t)}$ on $\mathcal{D}^{(t-1)}$
 - 4: Select next query point:
 - 5: $\mathbf{x}^{(t)} \in \arg \max_{\mathbf{x} \in \mathcal{S}} u(\mathbf{x}; \mathcal{D}^{(t-1)}, \hat{f}_1^{(t)}, \dots, \hat{f}_m^{(t)})$
 - 6: Query $f(\mathbf{x}^{(t)})$
 - 7: Update data: $\mathcal{D}^{(t)} \leftarrow \mathcal{D}^{(t-1)} \cup \{\langle \mathbf{x}^{(t)}, f(\mathbf{x}^{(t)}) \rangle\}$
 - 8: **end for**
-

